

## Prompt 1 Initial Multimodal Screening

Continue Parkinson's Disease Drug Repurposing

CURRENT STAGE

Do NOT start Deep Validation.

We currently have 27 raw candidate outputs.

Screen the existing candidates only.

ALLOWED DATA SOURCES

Use only:

Monarch Initiative KG

PD-related biological associations

biological route / context

PubMed

exact-compound pharmacology

mechanism

functional phenotype

prior PD experimental evidence

contradictory evidence

CNS evidence

ClinicalTrials.gov

human PD exposure

clinical trials

Google Patents (patents.google.com)

patent prior-art / novelty screening ONLY

Justia Patents may only be used as a backup search/listing source.

Do not use Justia alone for a final patent-based decision.

Do not use PubChem patent pages, Espacenet, WIPO PATENTSCOPE,  
or USPTO search because they are unavailable in the current environment.

Do not use general WebSearch results, vendor pages, Wikipedia,  
blogs, or unsupported model knowledge as scientific evidence.

If evidence cannot be verified, report UNRESOLVED.

=====

STEP 1 — IDENTITY

=====

Resolve the exact candidate identity first.

Record:

candidate

canonical name

synonyms

discovery branch

target/action if supported

Merge duplicate exact compounds but preserve all discovery branches.

All classifications must concern the EXACT compound,  
not merely its target, pathway, drug class, or related molecule.

=====

STEP 2 — NOVELTY / PRIOR PD EXPOSURE

=====

Apply N rules deterministically and in order.

N0:

Exact compound is an established/approved PD treatment.

-> REJECT

N1:

Exact compound has been directly tested in:

human PD study

PD clinical trial

animal PD model

cellular PD model

PD-associated genetic/molecular experimental model

-> REJECT

N2:

No direct PD experiment, but exact compound has already been:

explicitly proposed as a PD therapeutic

explicitly claimed/included for PD in a therapeutic patent

substantially discussed/predicted as a PD repurposing candidate

-> REJECT from strict novel pool

N3:

No direct exact-compound PD evidence found, but its known

target/pathway has a clear PD biological connection.

-> KEEP

N4:

No meaningful exact Drug-PD connection found and the proposed

Drug-PD hypothesis arises mainly by combining distributed evidence.

-> KEEP

Assign Prior PD Exposure:

P0 = approved / established PD use

P1 = human PD study or clinical trial

P2 = animal / cell / PD-associated experimental model

P3 = explicit proposal / patent / computational prediction

P4 = no meaningful exact Drug-PD link found

IMPORTANT:

If N0, N1, or N2 is confirmed, stop expensive screening for that candidate and mark REJECT.

Patent rule:

If Google Patents confirms that the exact compound, verified synonym, or clearly corresponding prodrug/form is explicitly proposed or claimed for Parkinson's disease therapy:

N = N2

P = P3

Decision = REJECT

Target-class or related-compound patents are NOT sufficient.

=====

STEP 3 — M / F / DIRECTION / CNS

=====

Run this step only for candidates surviving N/P screening.

M — Mechanistic Evidence

M1 = weak / largely speculative mechanistic evidence

M2 = moderate mechanistic component evidence

M3 = strong mechanistic component evidence

Evaluate evidence supporting:

Exact Drug

→ target / process / phenotype

→ PD-relevant biology

Do not silently attribute target-class evidence to the exact compound.

F — Exact-Compound Functional Evidence

F0:

Only binding, target annotation, or mechanism-of-action evidence.

F1:

Exact compound experimentally modulates the relevant target/pathway, but no relevant downstream functional phenotype is demonstrated.

F2:

Exact compound produces a relevant functional phenotype in cell or animal experiments outside PD.

Examples:

mitochondrial function, lysosomal function, autophagic flux, inflammatory state, proteostasis, neuronal survival, oxidative stress, synaptic phenotype, cellular trafficking.

F3:

Exact compound demonstrates the relevant functional phenotype in a strong translational or human context while still lacking a direct PD therapeutic link.

DIRECTION

consistent:

drug action is compatible with correcting the supported PD-relevant biology.

inconsistent:

drug action opposes the better-supported biological direction.

unclear:

evidence is insufficient or contradictory.

Do not infer therapeutic direction from a generic association.

CNS

Classify:

feasible

uncertain

fatal\_liability

Absence of BBB/CNS evidence alone = uncertain.

Use fatal\_liability only when evidence gives a strong reason that the proposed CNS mechanism is not realistically achievable.

#### EVIDENCE RULE

For major claims, record the supporting source:

Monarch relationship

PMID

NCT ID

Google Patent number

Use these evidence labels where appropriate:

OBSERVED\_KG

OBSERVED\_LITERATURE

SUPPORTED\_CONTEXT

AGENT\_INFERENCE

CONTRADICTED

UNRESOLVED

Never present AGENT\_INFERENCE as an observed fact.

#### PRIORITIZATION

Preferred Round 3 profile:

P4

AND M2 or M3

AND F2 or F3

AND Direction = consistent

AND CNS != fatal\_liability

AND no direct PD clinical, animal, cellular,  
or PD-associated experimental exposure.

F0 candidates should normally not proceed.

F1 candidates may remain HOLD / lower priority.

Do not keep weak candidates merely to reach a target number.

#### OUTPUT

Return ONE complete screening table containing:

Candidate

Canonical identity

Discovery branch

Target/action

Exact prior PD evidence

Evidence source

N

P

M

F

Direction  
 CNS  
 Exact-compound functional phenotype  
 PD relevance of phenotype  
 Contradictory evidence  
 KEEP / REJECT / HOLD  
 Reason  
 Most important remaining inference gap  
 Then report:  
  
 total screened  
  
 REJECT count  
  
 HOLD count  
  
 KEEP count  
  
 ranked shortlist of strongest survivors  
 Do NOT start Deep Validation.  
 STOP after returning the screening table for review.

## Prompt 2: N/P Critic Agent

Do NOT rerun the full screening. Do NOT change existing M / F / Direction / CNS classifications. Do NOT start Deep Validation yet. Only re-check the N/P novelty status of the 8 current KEEP candidates: - Oltipraz - Ru265 - Neftamapimod (VX-745) - Emricasan (IDN-6556) - Bis-T-23 - ANX005 - PMX205 - Ruxolitinib For each exact compound: 1. Re-check prior PD exposure using: - PubMed - ClinicalTrials.gov - Google Patents 2. Google Patents is the primary patent-content source. Justia may only be used to locate a patent number and is not sufficient alone for an N2/P3 decision. 3. Apply the existing novelty rules: - N0/P0 = established PD use → REJECT - N1/P1 or P2 = direct human/clinical or experimental PD exposure → REJECT - N2/P3 = explicit PD therapeutic proposal/patent/prediction → REJECT - N3/N4 with P4 = no meaningful exact Drug-PD prior exposure found → retain 4. If Google Patents or ClinicalTrials.gov cannot be successfully checked, do NOT claim: "no prior PD evidence/patent exists." Instead record that component as: UNRESOLVED or SOURCE\_ACCESS\_FAILED. 5. Preserve all previously established: - M - F - Direction - CNS - functional evidence - mechanistic evidence Return a compact table with: Candidate | Previous N/P | PubMed check | ClinicalTrials.gov check | Google Patents check | Revised N/P | KEEP/REJECT/HOLD | Reason Then provide the final shortlist. STOP after the table. Do not start Mechanism Proposing or Deep Validation. Make sure the table is in excel file type

## Prompt 3: Mechanism Hypothesis Proposing and Independent Validation

INPUT CONTEXT  
 Use the finalized screening and subsequent N/P critic output as the starting context for Deep Validation.  
 For the candidates, preserve and reuse all available prior information, including:  
 canonical identity  
 discovery branch  
 previous N/P classification  
 previous M/F/Direction/CNS classification  
 previously retrieved evidence  
 known contradictory evidence  
 unresolved source-access issues  
 previous KEEP rationale  
 Do not restart candidate discovery or screening from zero.  
 However, previous screening results are not treated as final mechanistic proof.  
 Deep Validation must independently verify the relationships needed for the mechanism using the approved sources.  
 If newly retrieved evidence contradicts or changes a previous screening conclusion, explicitly record the conflict and explain the revision.  
 Do not silently overwrite prior results.  
 Perform Deep Validation for candidates:

Oltipraz  
PMX205  
Neflamapimod (VX-745)  
Emricasan (IDN-6556)  
ANX005  
Bis-T-23

Do not investigate any other candidates.  
Complete all candidates without waiting for further user confirmation.

=====

#### DATA SOURCES

=====

Use only:

PubMed  
Monarch Initiative  
ClinicalTrials.gov

Do not use any other source.

For PubMed:

Use title, abstract, PMID, and publication metadata first.  
Full text may be used only when the abstract is insufficient to verify a critical relationship.  
If full text is used, explicitly mark the evidence as FULL\_TEXT.

Do not use model background knowledge as evidence.

If an approved source cannot be accessed, record:

SOURCE\_ACCESS\_FAILED

If a relevant question cannot be resolved from the approved evidence, record:

UNRESOLVED

Do not substitute another source.

A source-access failure for one candidate or one relationship must not stop the overall analysis.

Continue with the available evidence and continue to the next candidate.

=====

#### TASK 1 — MECHANISM HYPOTHESIS PROPOSING

=====

For each candidate, independently explore and propose the strongest evidence-grounded mechanistic hypothesis connecting the exact compound to Parkinson's-disease-relevant biology.

Do not assume a predefined mechanism.

Do not require any specific intermediate entity type.

Do not require a fixed number of intermediate steps.

Allow the mechanism to emerge from the evidence.

The Agent must decide:

which biological relationships are relevant,  
which relationships should be investigated further,  
how the evidence can be connected,  
and where inference is necessary.

Prefer mechanisms that are biologically coherent, well supported, and require fewer unsupported assumptions.

Do not attempt to preserve the previous screening mechanism if the evidence supports a different mechanism.

If the strongest evidence requires revising the original mechanism, explicitly report the revision.

=====

#### EDGE-LEVEL EVIDENCE REQUIREMENT

=====

Every relationship in the final mechanism chain must be individually documented.

For every edge report:

Edge number  
Subject  
Predicate  
Object  
Evidence status  
Source

Source identifier  
Evidence location  
Brief evidence statement  
Evidence status must be one of:  
OBSERVED\_KG  
OBSERVED\_LITERATURE  
SUPPORTED\_CONTEXT  
AGENT\_INFERENCE  
CONTRADICTED  
UNRESOLVED  
Source identifier must be:  
PMID for PubMed  
Monarch identifier for Monarch  
NCT number for ClinicalTrials.gov  
Evidence location must indicate:  
ABSTRACT  
FULL\_TEXT  
MONARCH\_KG  
CLINICALTRIALS\_RECORD  
If an edge is supported by direct evidence, cite the exact source.  
If an edge is AGENT\_INFERENCE, do not provide a false citation for the inferred relationship.  
Instead, explicitly provide:  
the reasoning for the inference  
which observed/supported edges the inference is based on  
why those pieces of evidence justify proposing the bridge  
Never present AGENT\_INFERENCE as an established fact.  
Never silently convert association into causation.  
Never silently attribute target-class, pathway-class, phenotype-level, or contextual evidence to the exact compound.  
A failure to find evidence is not itself contradictory evidence.  
Do not label a zero-result search as CONTRADICTED.  
Use UNRESOLVED or explicitly state that no direct evidence was identified using the approved searches.  
=====

**TASK 2 — INDEPENDENT VALIDATION**  
=====

After Mechanism Hypothesis Proposing, run a separate independent Critic for each candidate.  
The Critic must actively test whether the proposed mechanism is weakened or invalidated by available evidence.  
The Critic must independently search the same three approved sources.  
It should examine contradictions, incompatible directions, context mismatches, reachability issues, alternative explanations, unsupported edges, and any other evidence that materially challenges the proposed hypothesis.  
Do not constrain the Critic to a predefined checklist if other important problems emerge.  
Every criticism must also include its evidence source.  
If the criticism is an inference rather than a directly observed fact, clearly state the reasoning.  
The Critic must not try to rescue a hypothesis when material contradictory evidence is found.  
=====

**FINAL OUTPUT**  
=====

For each candidate provide:  
Candidate identity  
Proposed mechanistic hypothesis  
Complete mechanism chain  
Edge-level Evidence Ledger  
Agent-inferred bridges and their reasoning  
Unresolved relationships  
Contradictory evidence  
Independent Critic findings  
Main remaining evidence gap

Explicit conflicts or revisions relative to the previous screening and N/P critic assessment

Final verdict

Final verdict must be one of:

HYPOTHESIS\_HOLDS\_UP

INCONCLUSIVE

HYPOTHESIS\_UNDERMINED

Brief justification for the verdict

Do not make therapeutic efficacy claims.

The goal is to produce a testable, evidence-grounded mechanistic hypothesis.

=====

EXECUTION RULE

=====

Complete Deep Validation for all candidates:

Oltipraz

PMX205

Neflamapimod

Emricasan

ANX005

Bis-T-23

Mechanism Hypothesis Proposing

→ Edge-level Evidence Ledger

→ Independent Critic

→ Final Verdict